OSTEOPOROSIS DETECTION

Using Deep Learning Techniques

PROBLEM DOMAIN & **MOTIVATION**

- Osteoporosis is a progressive and asymptomatic bone disease that weakens bones and increases fracture risk.
- It often remains undetected until fractures occur, leading to disability and reduced quality of life.
- DEXA scans, the standard diagnostic method, are costly, time-consuming, and limited in availability.
- Knee X-rays are routinely available in hospitals but are not effectively used for osteoporosis screening.
- Manual interpretation of X-rays is subjective and error-prone.
- There is a strong need for an automated, low-cost, and accurate screening approach.
- Deep learning enables learning complex bone patterns from medical images.

- Osteoporosis incidence is rapidly increasing due to aging populations, especially in developing countries.
- Many outpatient settings lack access to specialized bone density equipment.
- Early-stage conditions like osteopenia are difficult to identify using conventional visual inspection.
- Delayed diagnosis results in higher treatment costs and irreversible bone damage.
- Machine learning models can assist clinicians as decision-support tools, not replacements.
- Automated screening can help in large-scale population health monitoring.
- Lack of explainability in AI systems reduces clinical trust and adoption.

EXISTING SYSTEM— ANALYSIS

- DEXA Scan is the current gold standard for osteoporosis diagnosis.
- It measures bone mineral density (BMD) but requires specialized equipment.
- DEXA machines are expensive and not portable, limiting accessibility.
- Diagnosis often occurs only after significant bone loss.
- Traditional radiographic analysis depends on manual interpretation by experts.
- Manual assessment is time-consuming, subjective, and prone to inter-observer variability.
- Conventional computer-aided systems rely on handcrafted features, limiting accuracy.
- Most existing ML models focus only on binary classification (normal vs osteoporosis).

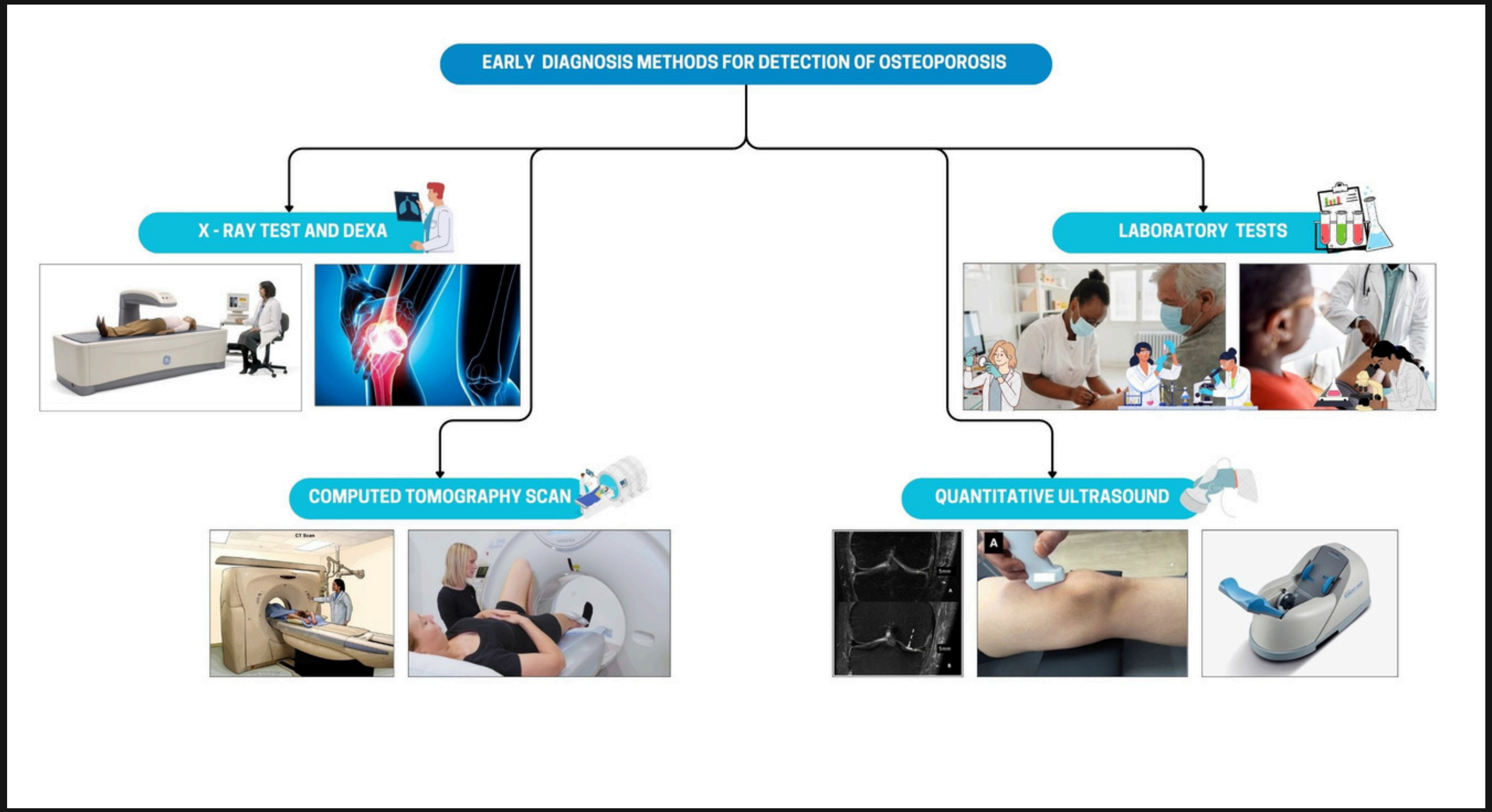
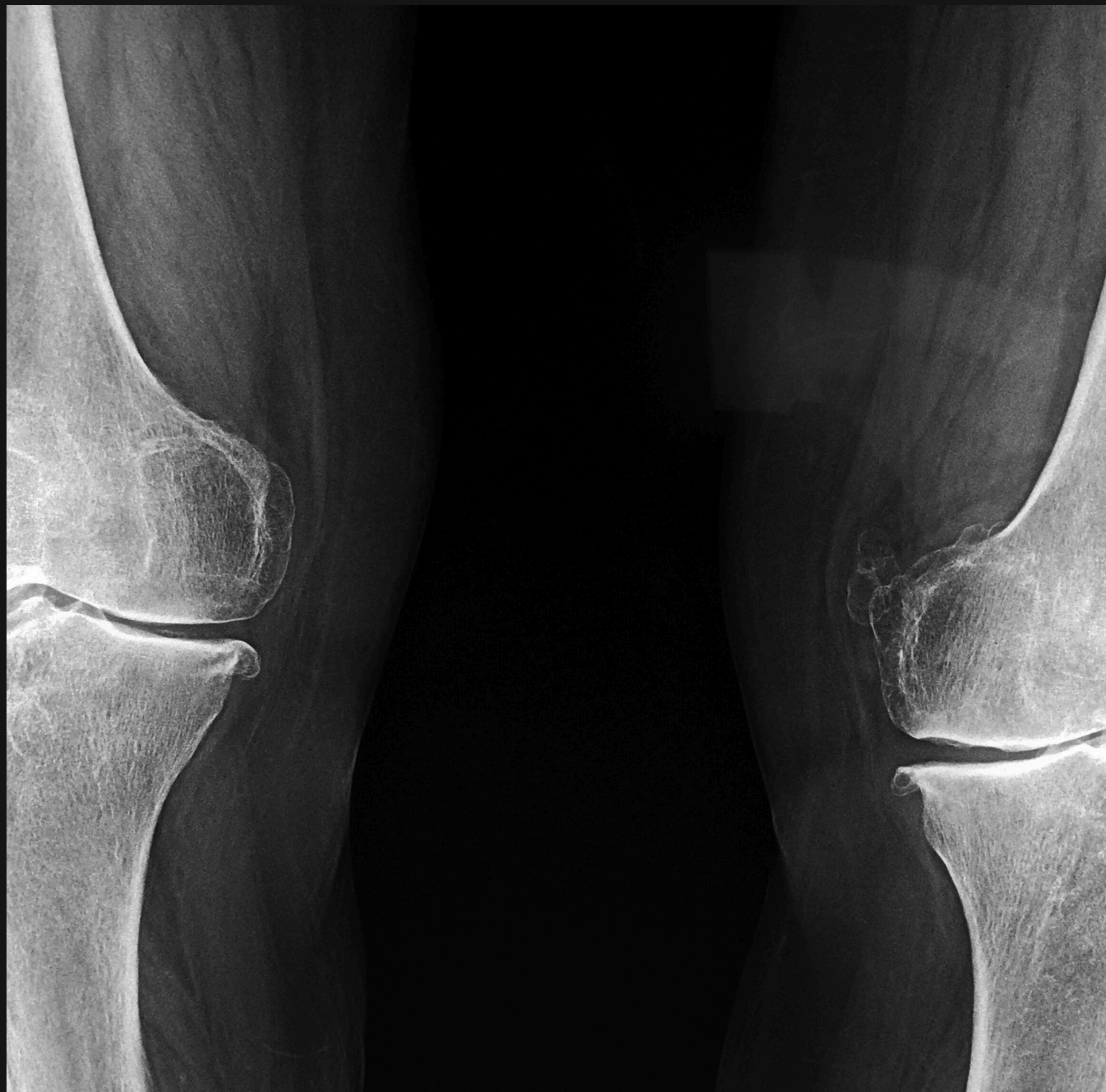


Figure 1: *Existing Methodologies*



FEASIBILITY STUDY AND LIMITATIONS OF EXISTING SYSTEMS

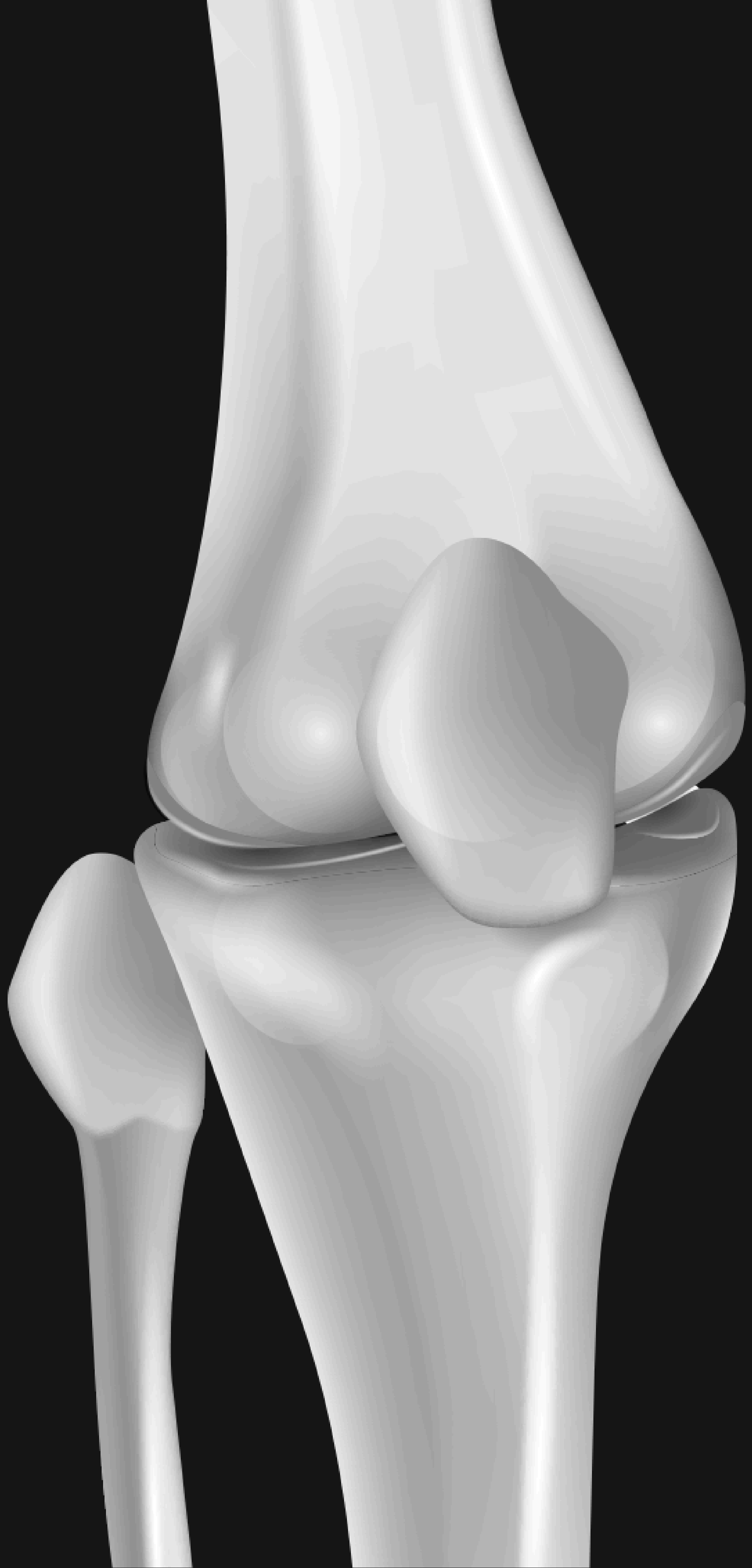
LIMITATIONS OF EXISTING SYSTEMS

- High dependency on DEXA equipment, which is costly and infrastructure-intensive.
- Limited availability of DEXA scanners in rural and semi-urban regions.
- Late diagnosis due to absence of routine screening mechanisms.
- Manual X-ray analysis suffers from subjectivity and human error.
- Traditional CAD systems show poor generalization on unseen datasets.

FEASIBILITY OF PROPOSED SYSTEM

- Knee X-ray images are already available in routine clinical practice.
- Transfer learning enables high performance with limited medical data.
- GPU-based training makes the system computationally feasible.
- Deployment through Streamlit allows real-time and user-friendly access.
- Explainable AI improves clinical acceptance and trust.

“The proposed system is economically, technically, and operationally feasible using existing radiology infrastructure.”



OBJECTIVES

- To develop a deep learning–based system for automated osteoporosis detection using knee X-ray images.
- To apply transfer learning techniques for improving classification accuracy on limited medical datasets.
- To perform multi-class classification into normal, osteopenia, and osteoporosis categories.
- To address class imbalance issues using proper data handling techniques.
- To integrate explainable AI methods such as SHAP or Grad-CAM to interpret model decisions.
- To evaluate the model using clinical performance metrics like precision, recall, F1-score, and ROC-AUC.
- To design a user-friendly interface for real-time screening support.

These objectives ensure not only high accuracy but also fairness, interpretability, and clinical usability of the system

METHODOLOGY

This pipeline ensures accurate, interpretable, and clinically meaningful osteoporosis detection from knee X-rays

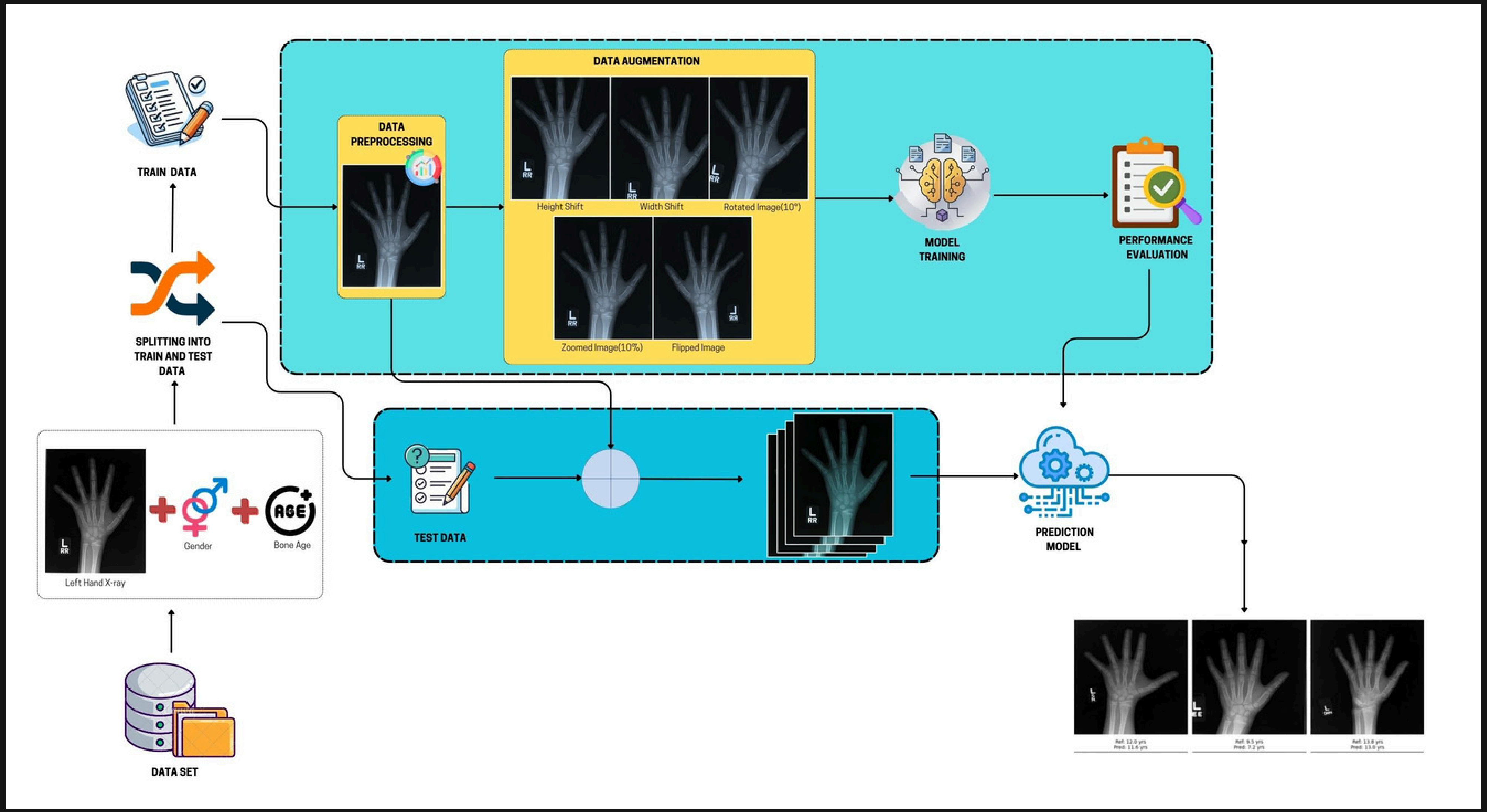
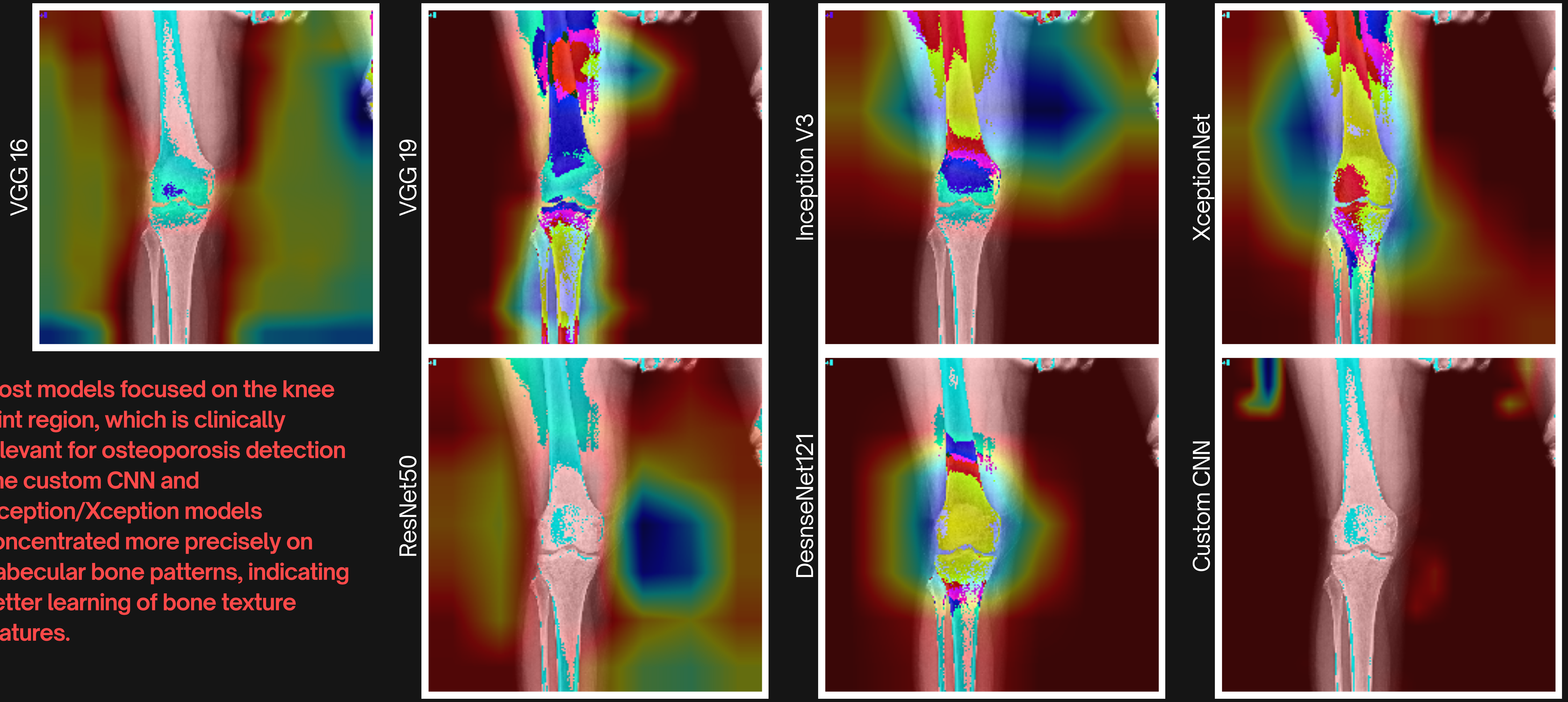


Figure 2: *Proposed Methodology*

DATASET

Dataset Source	Normal	Osteopenia	Osteoporosis	Type
Dataset 1 – Kaggle (Binary)	372	–	372	Binary
Dataset 2 – Kaggle (Binary)	186	–	186	Binary
Dataset 3 – Kaggle (Binary)	186	–	186	Binary
Dataset 4 – Mendeley (Multiclass)	36	154	Included (count not specified)	Multiclass
Dataset 5 – Using for this project	780	374	793	Multiclass

DATASET DESCRIPTION



- Most models focused on the knee joint region, which is clinically relevant for osteoporosis detection
- The custom CNN and Inception/Xception models concentrated more precisely on trabecular bone patterns, indicating better learning of bone texture features.

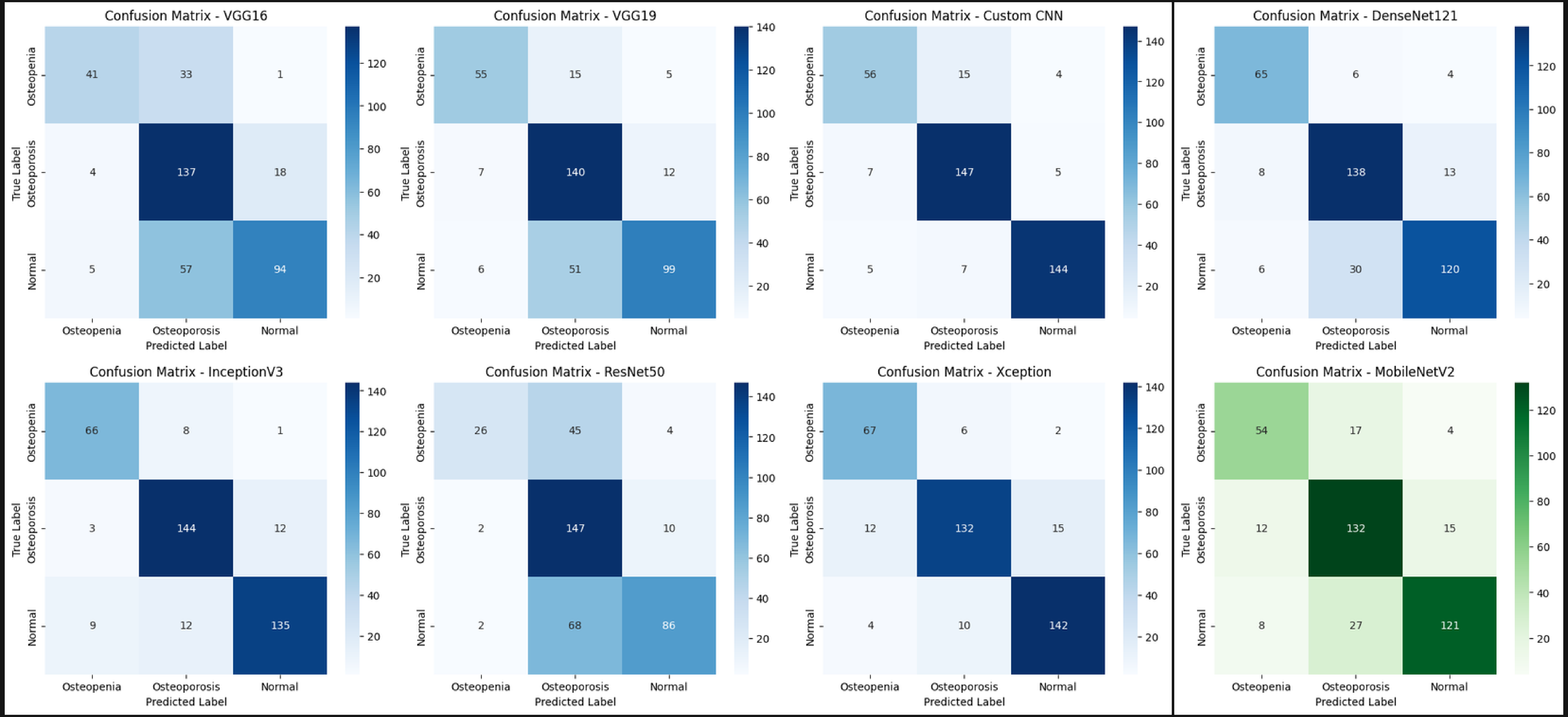
PROPOSED MODELS

- **Data Preprocessing**
  - All images were resized to a fixed input size required by CNN models
  - Normalization was applied to scale pixel values
- **Data augmentation techniques were used:**
  - Rotation
  - Horizontal flipping
  - Zooming
- **Augmentation helped in:**
  - Reducing overfitting
  - Handling class imbalance
  - Improving generalization

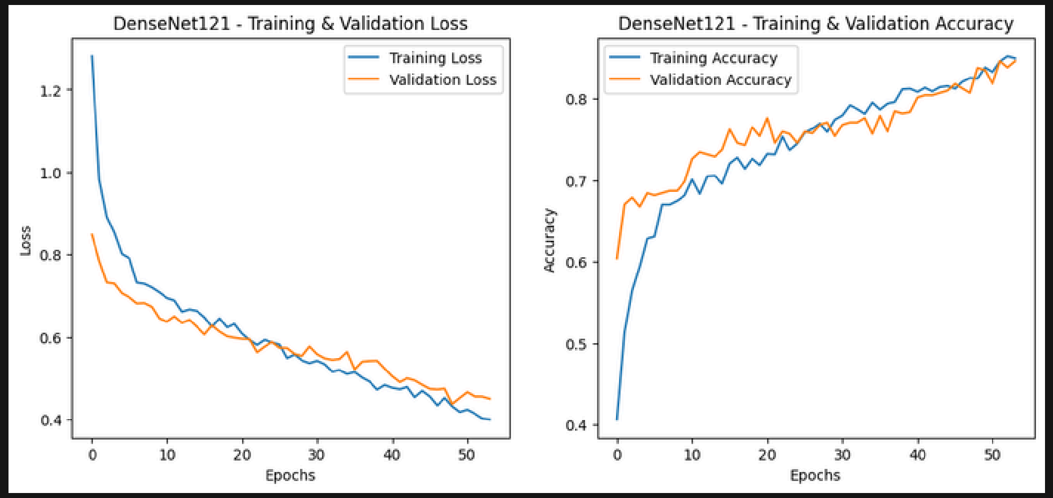
Model	Architecture Type	Key Strength	Computational Cost	Performance Trend
VGG16	Deep CNN (16 layers)	Simple & stable architecture	High	Good baseline accuracy
ResNet50	Residual Network	Handles deep networks efficiently using skip connections	Medium–High	Better than VGG
InceptionV3	Multi-scale CNN	Captures fine texture and spatial details	Medium	High accuracy
VGG19	Deep CNN (19 layers)	Rich feature extraction	High	Comparable / slightly better than VGG16
DenseNet121	Dense Connected Network	Reuses features from previous layers, improves gradient flow	Medium	Strong performance in medical imaging
MobileNetV2	Lightweight CNN	Designed for mobile & real-time applications	Low	Slightly lower but efficient accuracy
Xception	Depthwise Separable CNN	Efficient spatial feature extraction with fewer parameters	Medium	Near Inception-level accuracy
Custom CNN	Domain-specific CNN	Tailored for knee X-ray patterns	Low–Medium	Highest performance on our dataset

RESULTS

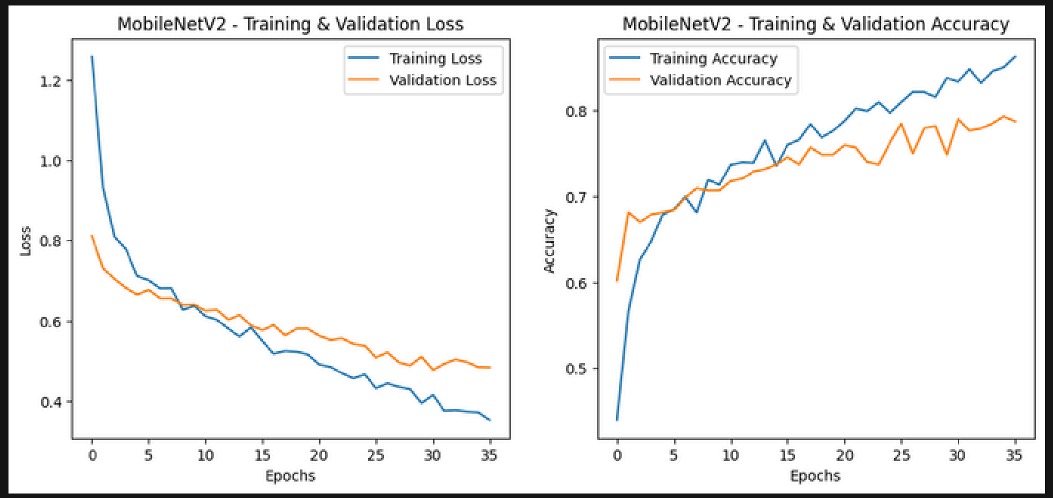
CONFUSION MATRICES



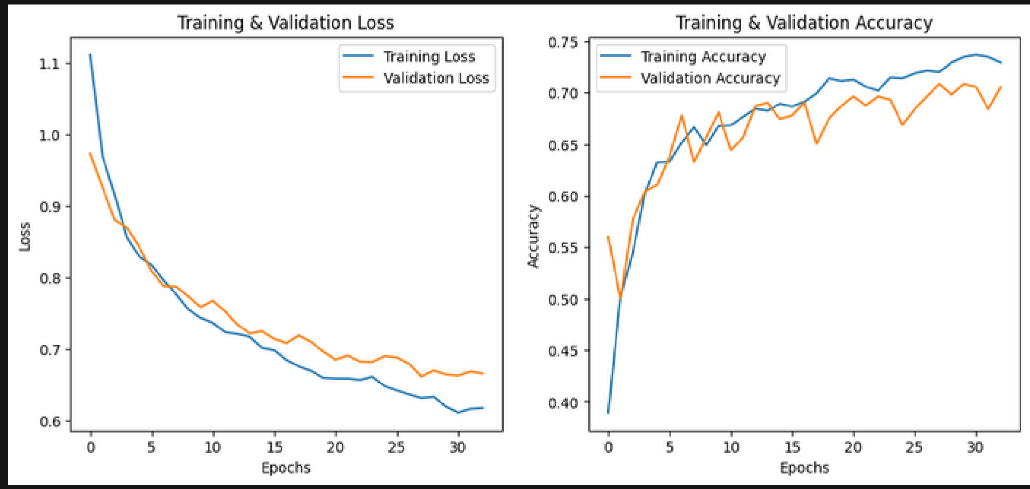
DenseNet121



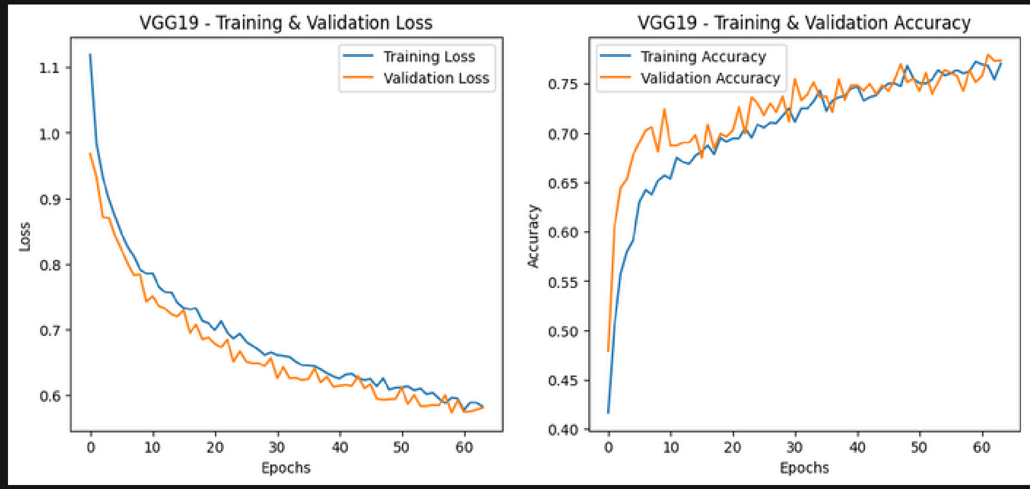
MobileNetV2



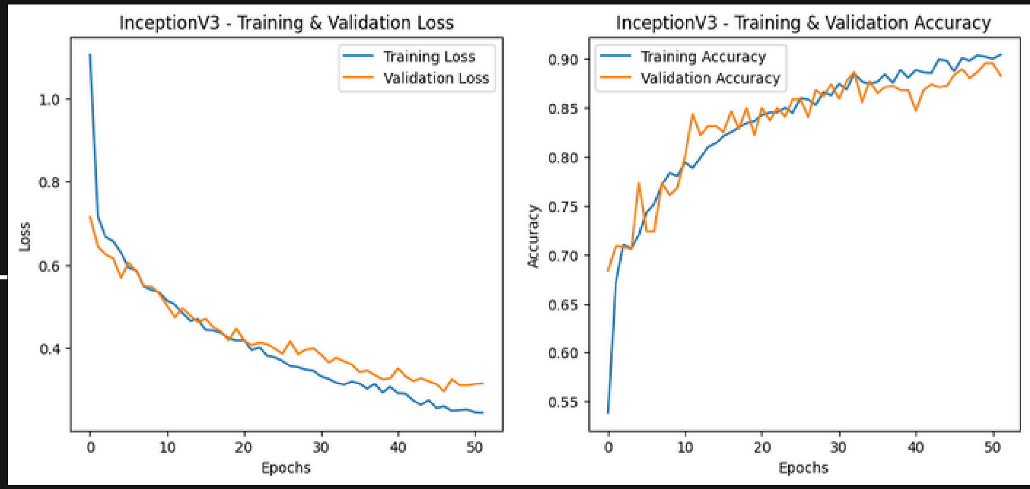
VGG 16



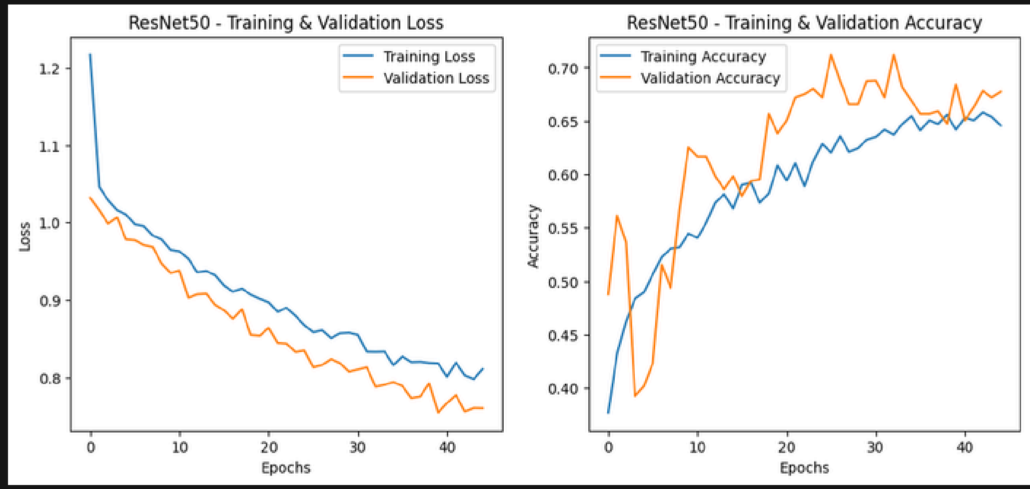
VGG 19



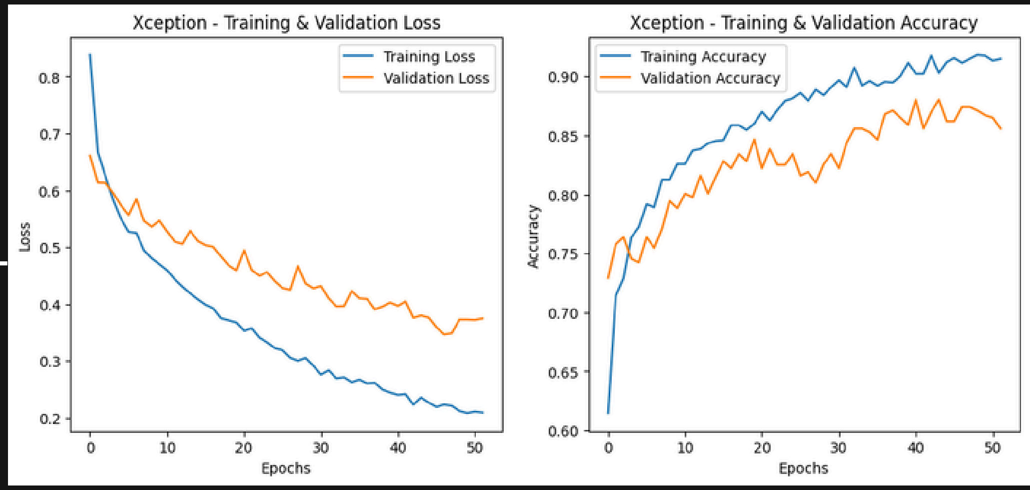
InceptionV3



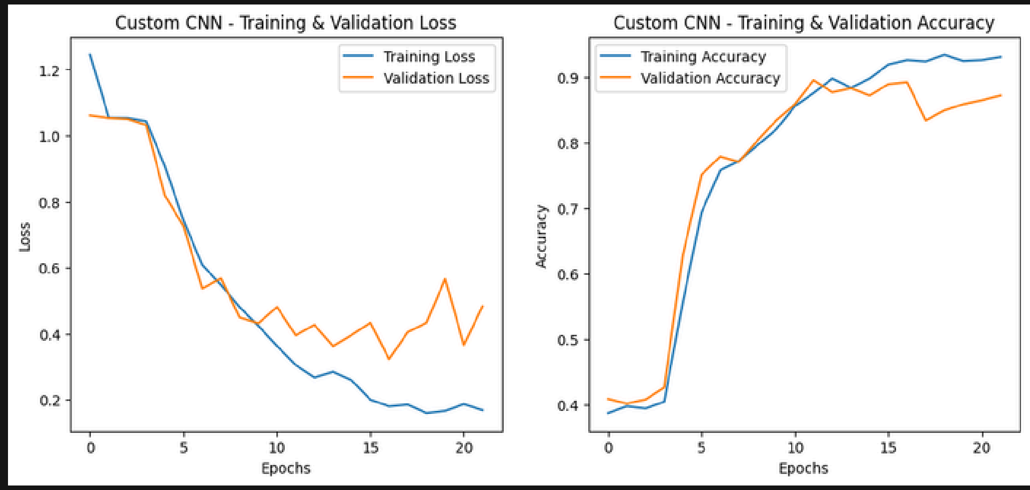
ResNet50



XceptionNet



Custom CNN

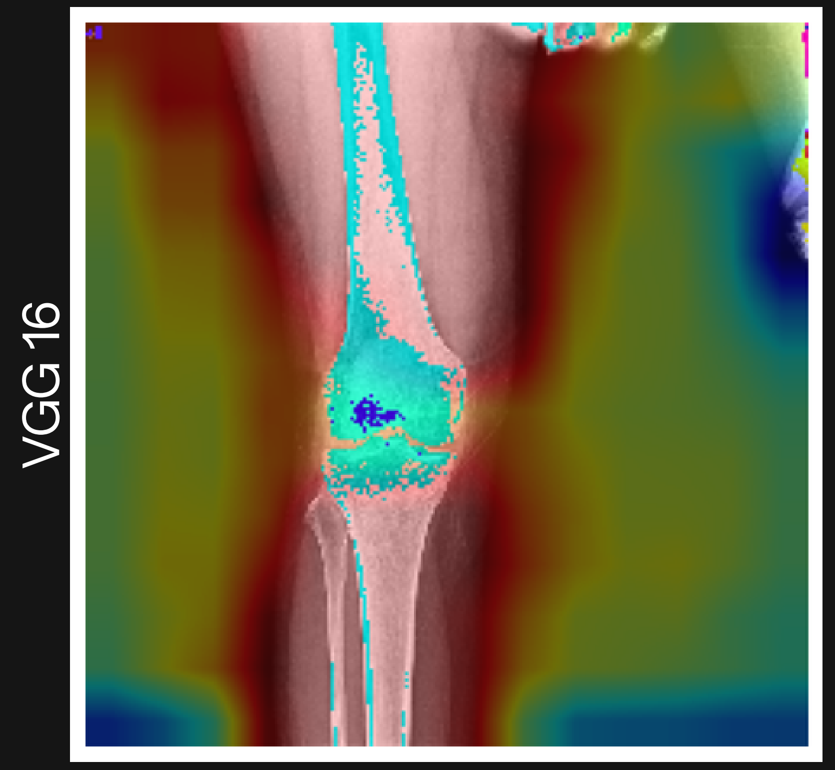


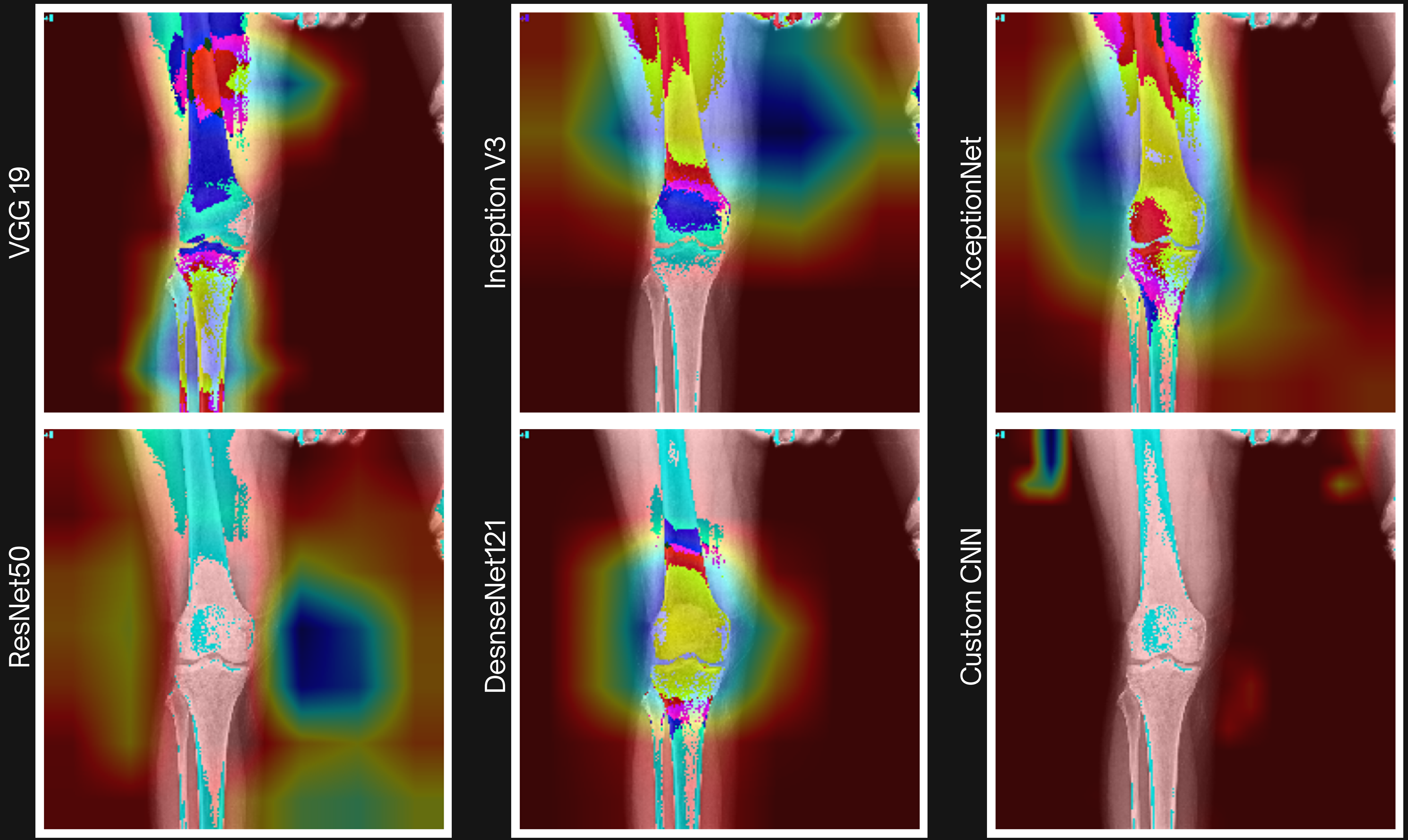
TRAINING GRAPHS

Model	Accuracy	Precision	Recall	F1-Score
VGG16	70%	0.74	0.7	0.7
VGG19	75%	0.77	0.75	0.75
ResNet50	66%	0.74	0.66	0.65
InceptionV3	88%	0.89	0.88	0.88
DenseNet121	83%	0.83	0.83	0.83
MobileNetV2	84%	0.82	0.84	0.82
Xception	87%	0.88	0.87	0.87
Custom CNN	**89%**	**0.89**	**0.89**	**0.89**

Table 2: *Comparison of Results*

GRAD-CAM RESULTS





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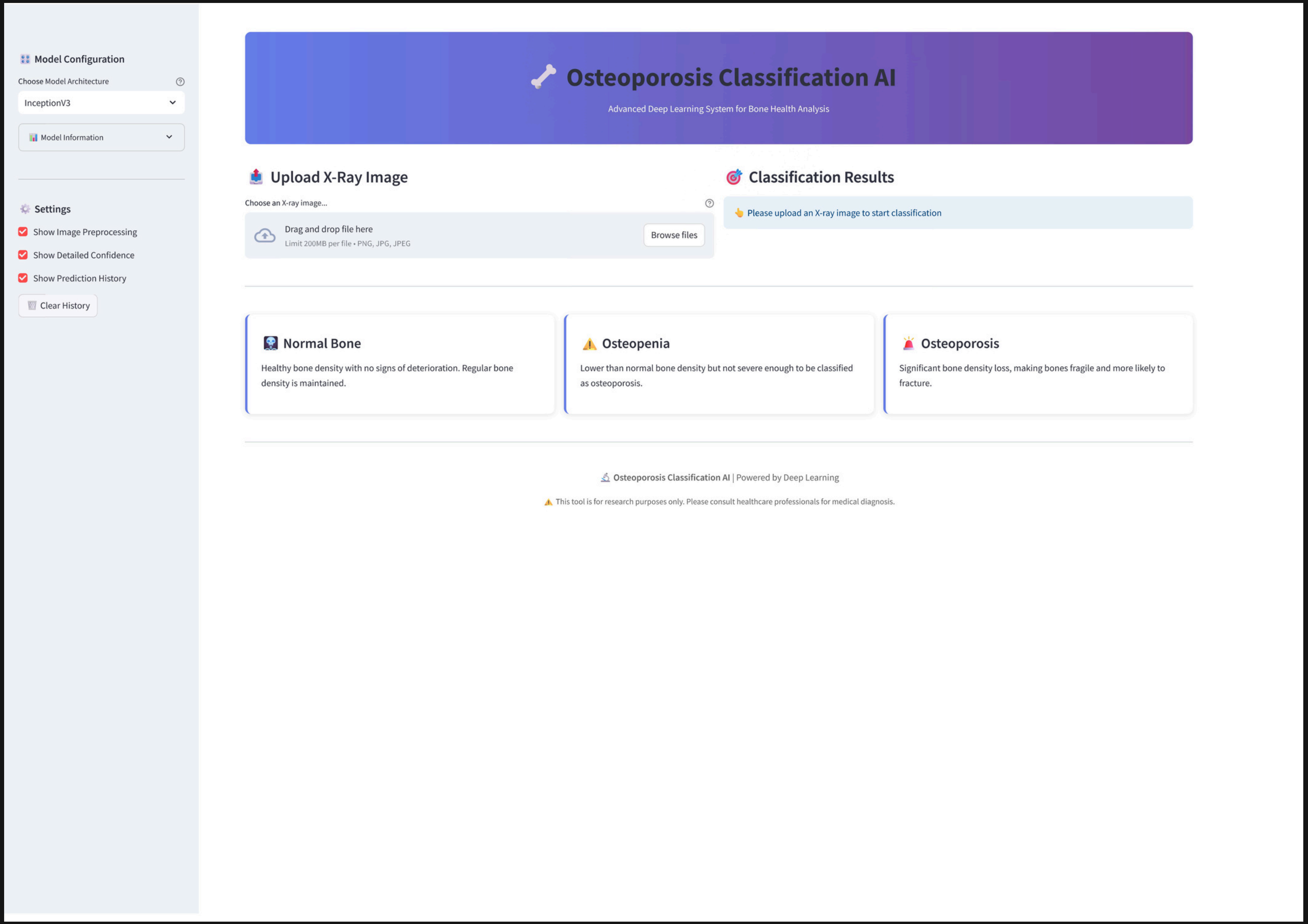
UI APPLICATION



CONCLUSION & DISCUSSION

- The project successfully developed a deep learning-based system for multiclass osteoporosis detection using knee X-ray images.
- Multiple CNN architectures were trained and evaluated.
- Among all models:
    - Custom CNN achieved the highest accuracy of 89%.
    - InceptionV3 achieved 88% accuracy.
    - Xception achieved 87% accuracy.

Grad-CAM explainability techniques were used to visualise the regions influencing model predictions.

The results demonstrate that deep learning can effectively support automated osteoporosis screening and clinical decision support.

FUTURE SCOPE

- Develop a more interactive web or mobile application to allow doctors and healthcare workers to easily upload X-ray images and obtain predictions.

1. User-Friendly Interface

- The current work can be extended into a full research publication to contribute to the field of AI-assisted medical diagnosis.

2. Research Paper Publication

- Collect larger knee X-ray datasets from Indian hospitals to improve model accuracy and ensure better generalization across different populations.

3. Dataset Collection from India

- Combine predictions from multiple deep learning models using ensemble techniques to further improve classification accuracy and robustness.

4. Integration of Ensemble Models

THANK YOU